

4 stages (encoder)
input (1, 192, 192)

stage 1

entry cov $\rightarrow$ (64, 192, 192)

res $\times 2 \rightarrow$ (64, 192, 192)

downsample $\rightarrow$ (128, 96, 96)

stride (2, 2, 2)
 $\uparrow$

3 times
downsample

num-res-unit

$\uparrow$
each layer =

2 res units

stage 1

entry conv $\rightarrow$ (128, 96, 96)

res x 2 $\rightarrow$ (128, 96, 96)

downsample $\rightarrow$ (256, 48, 48)

stage 3 same prev (l^t)

$\rightarrow$ (512, 24, 24)

stage 4 entry conv + res x 2

Skip connections

skip 1: (64, 192, 192)

skip 2: (128, 96, 96)

skip 3: (256, 48, 48)

output of
Layer 1

2

3

↓ (decoder)

512 x 24 x 24 Stage 4

Stage 5

upsample (256, 48, 48)

concatenation (256 + 256, 48, 48)

compressed conv (256, 48, 48)

res X2 (256, 48, 48)

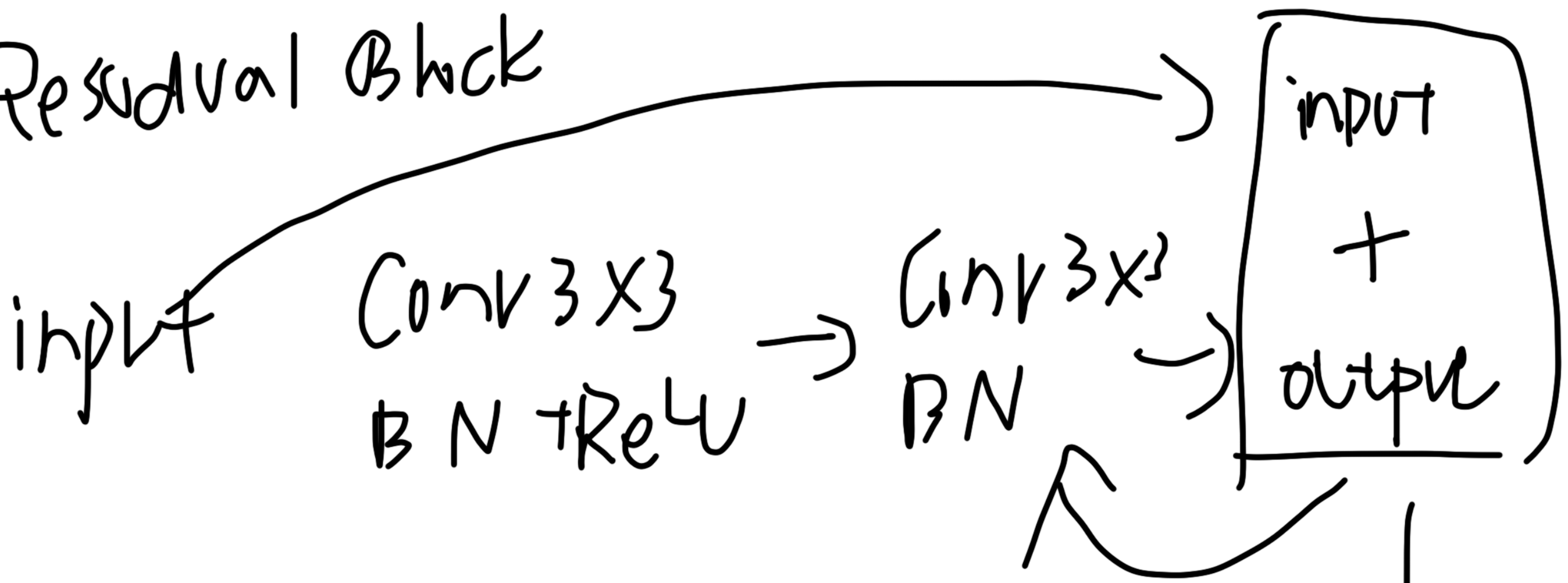
↓
stage 6, 7 same as stage 5
+ skip 2, 1

⇒ $64^x 192 \times 192$

output 1x1 conv (4, 192, 192)

for initial
softmax

Residual Block

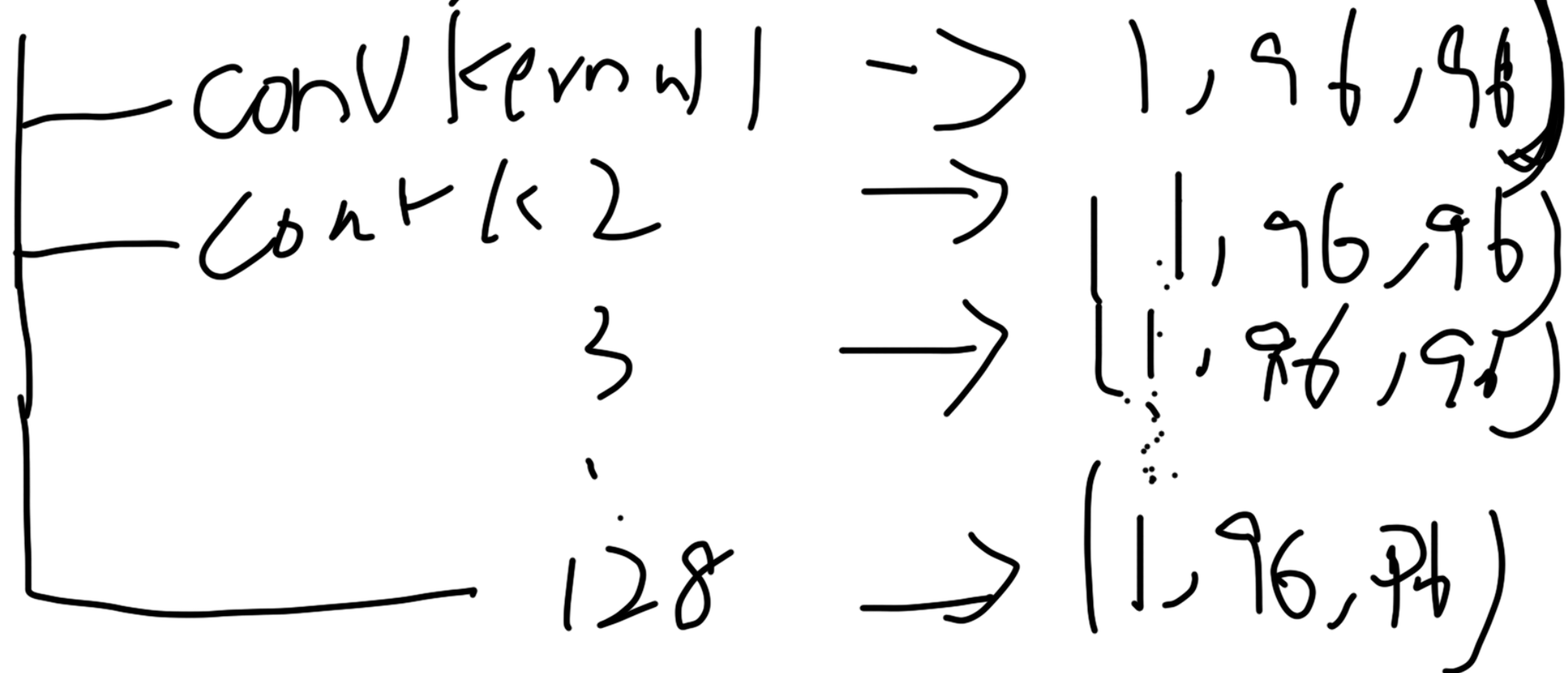


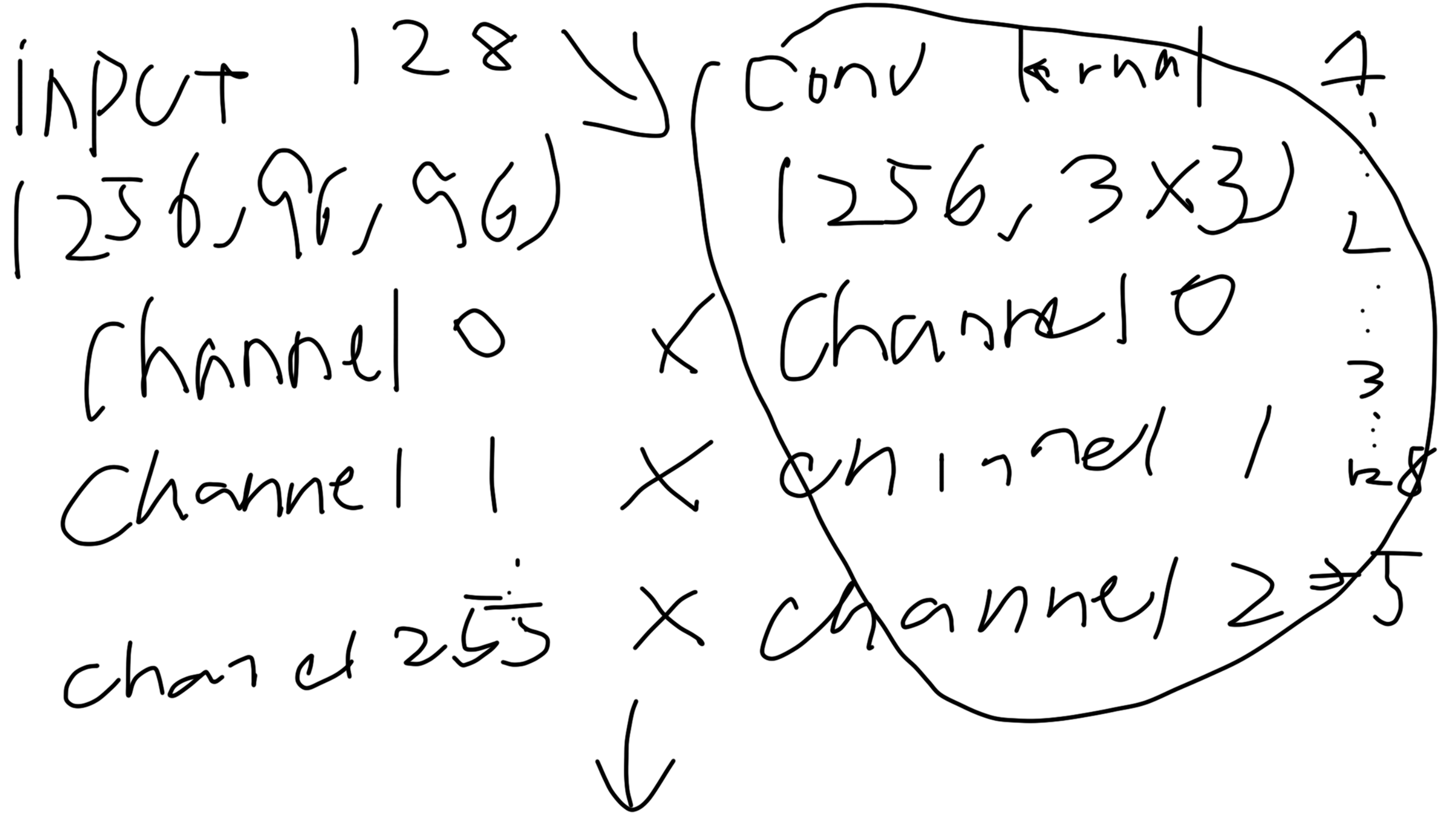
$$\text{output} = \text{ReLU}(x + \text{Conv}(x)) \text{ --- ReLU}$$

Compress conv

Assume

input (256, 96, 96)





The 3x3 area corresponding elements of all channels are multiplied and then added together